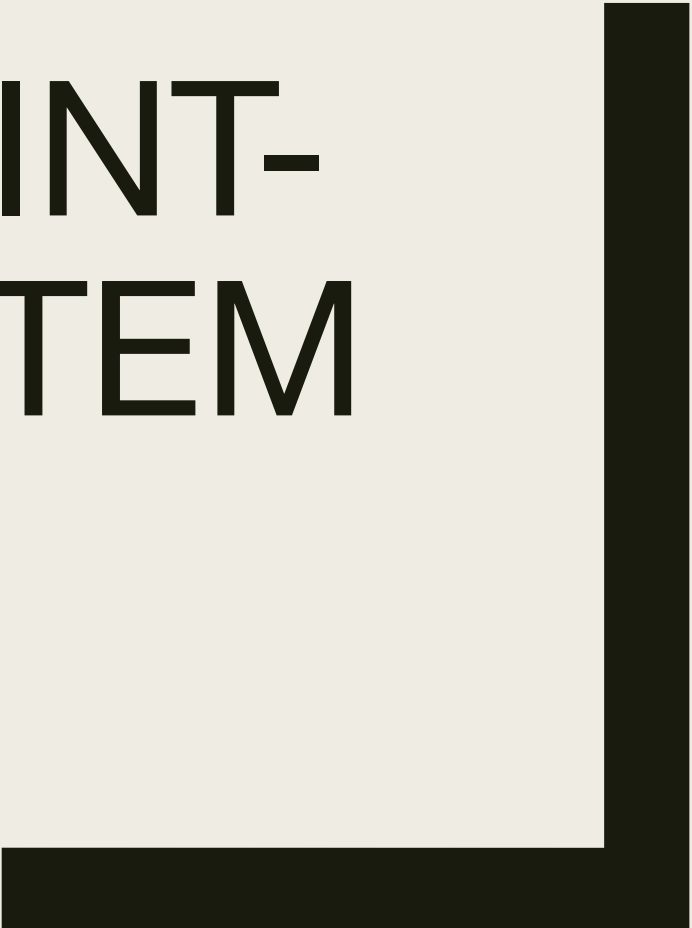




HOT DOG POINT- TO-SALE SYSTEM

Business Presentation
Yina Cai



Kata Overview

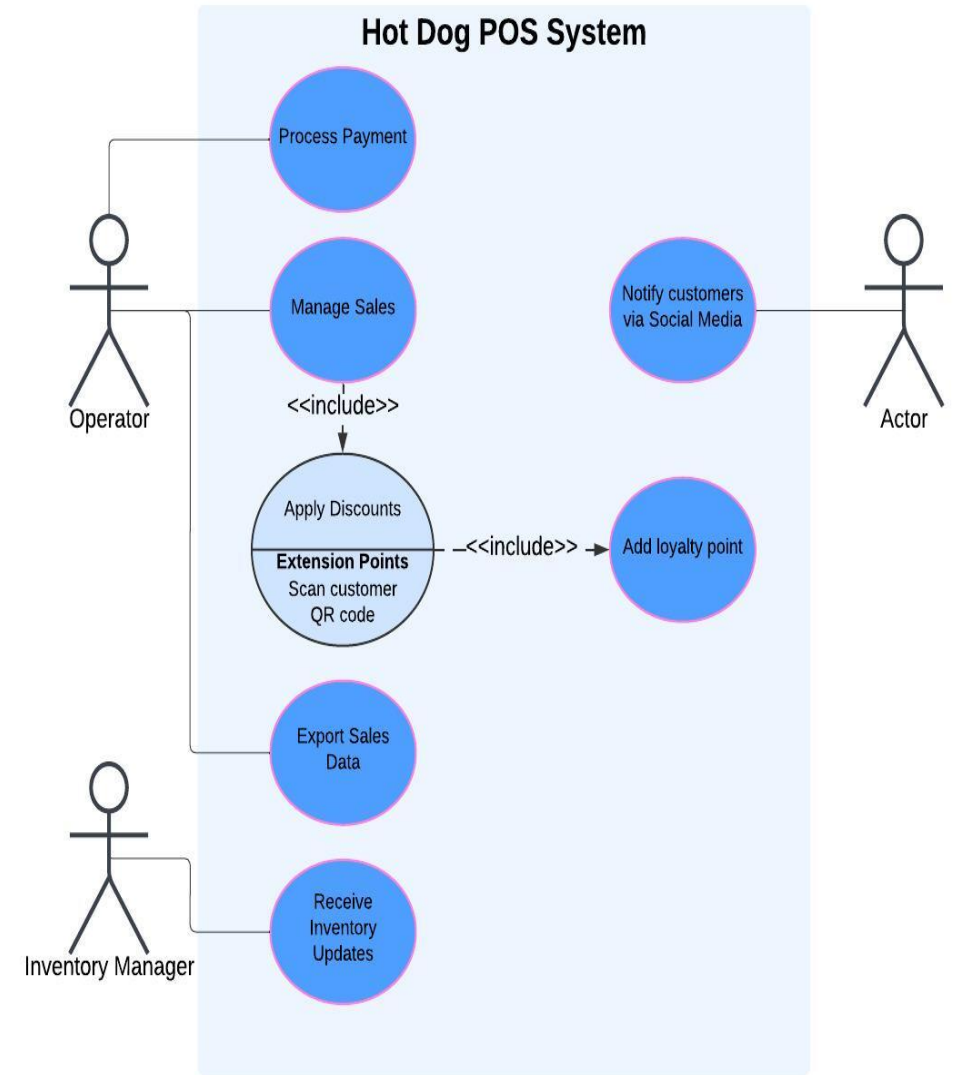
- The goal is to design a modern, lightweight, and scalable point-of-sale system for a local hot dog stand merchant. This system will support 50+ operators and serve thousands of customers, it offers features like discounts, sales tracking, inventory updates, social media integration, and exportable financial data. The purpose of this system is reducing manual effort. Key constraints are time-to-market urgency and the solution needs to be feasible for at least three years, but there are no serious budget constraints.

Architecturally Interesting Bits

- Lightweight operation. The system must run on small portable devices(Tables or smartphones).
- Real-time inventory updates.
- Social Media Integration.
- Scalability to suit for future business growth.

Proposed Solution

Microservices Architecture with Event-Driven Communication.



Failure Cases and Mitigation

- System overload: Auto-scale cloud infrastructure to handle increased traffic and. add resources dynamically.
- Service downtime: Deploy services across multiple regions with failover mechanisms.
- Data loss: Use event persistence and retry mechanisms in the event broker.

Business Drivers and Impacts

- Customer Engagement
- Customer Acquisition
- Operational Efficiency
- Future Scalability

Expected Data Growth and Impacts

- Sales Transactions
- Social Media API Usages
- Inventory Data

Business Continuity Needs

- High Availability
- Disaster Recovery
- Support and Maintenance